Retail Sales Performance Analysis

Blinkit Grocery Dataset – Power BI Dashboard Report

1. Introduction

This report presents a comprehensive analysis of Blinkit's retail sales data, visualized through a Power BI dashboard. The objective of the analysis is to explore patterns in sales performance, outlet characteristics, and product attributes to identify key factors influencing revenue generation.

The analysis focuses on three major areas:-

- **Overall business performance**
- **Outlet-level operational insights**
- **Product and pricing intelligence**

2. Objectives of the Analysis

The analysis aims to answer the following questions:

- Which product categories generate the highest revenue?
- Which outlets are the top contributors to overall sales?
- How do outlet size and location affect store performance?
- Does the age of an outlet influence its sales performance?
- How do product attributes such as weight, price, and fat content influence demand?
- Is there any relationship between product price and sales volume?

These questions guide the design of the dashboard and the analytical structure of the report.

3. Dataset Overview

The dataset represents transactional retail data collected from multiple Blinkit outlets.

The dataset contains information related to:

Product Attributes

- Item Identifier
- Item Type (Category)
- Item Weight
- Item Fat Content
- Item Visibility
- Item Maximum Retail Price (MRP)

Outlet Attributes

- Outlet Identifier
- Outlet Type
- Outlet Size
- Outlet Location Tier
- Outlet Establishment Year

Sales Information

- Item Outlet Sales (Revenue generated from the product in a specific outlet)

4. Data Modelling Approach

To support efficient analysis, the dataset was structured using a **Star Schema data model**.

The model consists of:

Fact Table

Fact_sales

Contains transactional sales data including product identifiers, outlet identifiers, and sales values.

Dimension Tables

Dim_product
Contains descriptive product attributes such as item type, weight, fat content, and price.

Dim_outlet
Contains outlet-related attributes including outlet size, outlet type, location tier, and establishment year.

This modelling approach ensures efficient filtering and aggregation while maintaining a clear separation between **transactional data and descriptive attributes**.

5. Analytical Methodology

The dashboard was designed using three analytical layers:

1. **Executive Overview**
2. **Outlet Performance Analysis**
3. **Product and Pricing Intelligence**

Each layer focuses on answering specific business questions and provides different perspectives on the dataset.

The analysis uses several calculated measures created using DAX (Data Analysis Expressions) to derive key metrics such as total sales, average pricing, and outlet efficiency.

5.1 Executive Overview Analysis

The Executive Overview page provides a high-level summary of Blinkit's overall retail performance.

Key performance indicators displayed include:

- Total Sales
- Sales per Outlet
- Average Product Price
- Number of Product Types
- Number of Outlets
- Number of Location Tiers

Category-Level Sales

The analysis shows that **fruits and vegetables and snack foods generate the highest total revenue**. These categories appear to be core contributors to the retail portfolio and represent strong customer demand.

Lower-performing categories such as seafood and certain specialty foods contribute a smaller portion of the overall revenue.

Outlet Type Performance

The analysis indicates that **Supermarket Type 1 outlets generate the largest share of total sales**. This suggests that this outlet format is likely the most effective retail model for Blinkit in terms of revenue generation.

Outlet Size Distribution

Sales performance varies significantly by outlet size. Medium-sized outlets appear to generate the highest total sales, indicating that store size may influence customer accessibility and product assortment.

Location-Based Sales

The analysis also shows that **Tier 3 locations contribute the highest share of revenue**. This insight may indicate strong demand in smaller cities or emerging markets.

Outlet Establishment Trends

Sales trends by outlet establishment year suggest that certain outlets experience different performance levels depending on when they were established. While older stores sometimes show strong sales performance, the relationship is not strictly linear.

5.2 Outlet Performance Analysis

The Outlet Performance page focuses on understanding **store-level efficiency and geographic performance differences**.

Top Performing Outlets

The analysis identifies a small group of outlets responsible for a large portion of the total sales. These outlets represent the highest-performing stores in the network.

Understanding what differentiates these outlets (location, size, or product mix) may help replicate their success in other stores.

Location Tier Performance

Sales across location tiers show that **Tier 3 locations outperform Tier 1 and Tier 2 locations**. This finding suggests that demand for grocery products may be stronger in developing markets.

Outlet Type and Location Interaction

The matrix analysis comparing outlet type and location tier shows that **Supermarket Type 1 performs consistently across multiple location tiers**, making it the most reliable outlet format.

Outlet Age vs Sales

A scatter analysis comparing outlet age and sales indicates that while some older outlets generate strong revenue, there is no strictly proportional relationship between outlet age and performance. This suggests that factors beyond store age—such as product mix or customer demand—play a significant role.

5.3 Product and Pricing Intelligence

The Product & Pricing Intelligence page focuses on understanding **product characteristics and pricing behaviour**.

Category Performance

The top-selling product categories include:

- Fruits and Vegetables
- Snack Foods
- Household Items

These categories dominate overall sales and represent the most significant contributors to revenue.

Product Weight Distribution

The weight distribution analysis shows that most products fall within **medium and heavy weight categories**, indicating the typical size range of products sold in the store.

Fat Content Analysis

An examination of fat content indicates that **low-fat products appear more frequently among lighter product categories**, while heavier products often belong to the regular-fat category.

This may reflect typical consumer product composition patterns.

Price vs Sales Relationship

A scatter analysis comparing product price and total sales suggests that **higher-priced products do not always generate higher sales**. Many moderately priced products outperform premium products in terms of demand.

This indicates that consumer purchasing behaviour is influenced by factors such as product necessity and category demand rather than price alone.

Weight vs Sales Relationship

The analysis comparing product weight and sales shows no strong linear relationship. While some heavier products generate strong revenue, weight alone does not appear to be a primary driver of demand.

Instead, product category and consumer preference appear to play a larger role.

6. Key Insights

Several important insights emerge from the analysis:

1. Supermarket Type 1 outlets dominate overall sales performance.
2. Medium-sized outlets appear to be the most effective store format.
3. Tier 3 locations generate higher revenue compared to Tier 1 and Tier 2 locations.
4. Fruits and vegetables and snack foods are the most important product categories.
5. Product price and product weight do not show a strong direct relationship with sales performance.

10. Conclusion

This analysis demonstrates how retail data can be transformed into actionable insights using Power BI. By combining outlet characteristics, product attributes, and sales data, the dashboard provides a holistic view of Blinkit's retail operations.

The results highlight the importance of **outlet format, location strategy, and product category** management in driving retail success.

Overall, the dashboard enables decision-makers to better understand **store performance**, identify **high-performing products**, and evaluate the effectiveness of different outlet types and locations.

Such insights can support more informed decisions regarding **store expansion, product assortment, and pricing strategy**.